

FTEC5660 Homework 2 – Part 2

Moltbook Social Agent

Autonomous AI Agent for a Reddit-like Social Network

Student ID: 1155249660 | Agent: ql_69204331 | Platform: Moltbook

1. Agent Design & Architecture

The Moltbook Social Agent is an autonomous LLM-powered agent that interacts with the Moltbook platform – a Reddit-like social network designed for AI agents. The agent uses DeepSeek-Chat as the reasoning core and a custom REST-based tool set wrapping the Moltbook API (v1).

Component	Detail
LLM Core	DeepSeek-Chat (temperature=0.3, max_tokens=2048)
API Layer	Moltbook REST API v1 at www.moltbook.com/api/v1
Authentication	Bearer token: moltbook_sk_w-my4MQDLnLyxuvvM5kWf68Fb3olwkp
Agent Identity	ql_69204331 (Student ID 1155249660 encoded via affine cipher)
Framework	LangChain 1.0 bind_tools() + synchronous tool dispatch loop
Verification	DeepSeek LLM-based obfuscated math challenge solver

Tool Set (12 tools)

Tool	Purpose
get_agent_profile	Read own Moltbook profile & claim status
check_claim_status	Verify human-claim linkage
subscribe_to_submolt	Join a submolt community
get_submolt_info	Fetch submolt metadata
list_submolts	List all available submolts
get_feed	Retrieve platform feed (hot/new/top/rising)
get_post	Fetch a single post by ID
upvote_post	Like / upvote a post
add_comment	Post a comment (may trigger verification)
submit_verification	Answer the obfuscated math challenge

Tool	Purpose
search_moltbook	Semantic search across posts & comments
get_post_comments	List comments on a specific post

2. Decision Logic & Degree of Autonomy

Task Execution Strategy

The agent uses a **direct API + LLM hybrid** strategy: critical tasks (subscribe, upvote, comment) are executed directly via Python requests for reliability, while the LLM agent is invoked for confirmation and status reporting. This hybrid approach avoids LLM hallucination on structured API calls.

Step	Task	Executor	Detail
1	Check Profile	Direct API	GET /agents/me → verify agent is authenticated
2	Subscribe /m/ftec5660	Direct API	POST /submolts/ftec5660/subscribe
3	Get Target Post	Direct API	GET /posts/{TARGET_POST_ID}
4	Upvote Post	Direct API	POST /posts/{id}/upvote
5	Comment on Post	Direct API	POST /posts/{id}/comments + verify
6	Status Verification	LLM Agent	Agent confirms all tasks via tool calls (max 8 turns)

Verification Challenge Handling

Moltbook requires solving an obfuscated math challenge before a comment is accepted. The challenge is a word-problem encoded with alternating caps and scattered punctuation (e.g., 'A] IO^bSt-Er S[wlmS aT/ tW]eNn-Tyy'). The agent:

- Detects **verification objects** in any API response (nested or top-level).
- Calls **solve_verification_challenge()** which sends the raw challenge text to DeepSeek.
- DeepSeek cleans the obfuscation, identifies the two numbers and operation, computes the result.
- Submits the answer formatted as 'X.XX' via **submit_verification()** tool.

Autonomy Level

- **High autonomy for structured tasks** – subscribe, upvote, comment execute without human confirmation.
- **LLM reasoning for ambiguous situations** – when verification challenges occur mid-task.
- **Graceful degradation** – each step is wrapped in try/except; failures are logged but do not abort subsequent steps.
- **Affine encoding of Student ID** – $E(x) = (11x + 10025379) \bmod 10^{14}$, ensuring unique agent identity.

3. Moltbook Interaction Log

The following table shows the key API interactions executed during the agent run:

Step	API Call	Status	Result
Step 1	GET /agents/me	✓ 200 OK	Agent ql_69204331 authenticated, claimed=True
Step 2	POST /submolts/ftec5660/subscribe	✓ 200 OK	Successfully subscribed to /m/ftec5660
Step 3	GET /posts/47ff50f3-...	✓ 200 OK	Target post retrieved; title confirmed
Step 4	POST /posts/.../upvote	✓ 200 OK	Post upvoted; vote count incremented
Step 5	POST /posts/.../comments	✓ 200 OK*	Comment submitted; verification challenge triggered & solved
Step 6	POST /verify	✓ 200 OK	Math answer accepted; comment published
Step 7	LLM Agent (8 turns)	✓ Completed	Agent confirmed all 5 tasks via tool calls

* Comment endpoint returns a verification challenge which is automatically resolved.

Comment Content Posted

"Hello from FTEC5660! This is an autonomous agent comment. I'm exploring the Moltbook platform as part of my coursework on Agentic AI for Business and FinTech. Excited to be part of this community!"

Verification Challenge Example (auto-solved)

Challenge: "A] l0^bSt-Er S[wImS aT/ tW]eNn-Tyy mE^tE[rS aNd] Sl0/wS bY^ fI[vE" →
Clean: "A lobster swims at twenty meters and slows by five" → Parse: 20 - 5 = 15 →
Submit: answer = "15.00" → Accepted ✓

4. Implementation Notes

Key Code Components

Function	Description
encode_student_id()	Affine cipher $E(x)=(11x+10025379) \bmod 10^4$
solve_verification_challenge()	LLM-based obfuscated math solver
run_agent()	LangChain LLM agent loop with bind_tools (up to 12 turns)
direct_*	Direct HTTP calls for subscribe/upvote/comment (reliability)
main()	Orchestrates 6 steps; falls back gracefully on any failure

Design Trade-offs

- **Direct API vs. LLM-mediated:** Direct calls are more reliable for deterministic tasks; LLM is used only where reasoning is needed.
- **Temperature 0.3 for comment:** Slightly higher than Part 1 to produce more natural social media language.
- **Auto-verification handler:** Embedded in both direct and LLM paths to ensure challenges are always resolved.